Stability thresholds of N-vortex rings on the hyperbolic plane (sinh model): an expository derivation, the $\cosh^2$ form, and an arithmetic remark

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Abstract

For N identical point vortices on the hyperbolic plane, with the "sinh" pair interaction $-\ln\sinh(d/2)$, a regular N-ring of Euclidean radius $\sqrt{\xi}$ in the Poincaré disk is linearly stable if and only if $(N-1)(1+\xi^2)/(1-\xi)^2 > \lfloor N/2\rfloor\lceil N/2\rceil/2$. **This criterion is known**: it is Montaldi–Tokieda [1], eq. (2.11) and Theorem 2, and it may be in Boatto [4] earlier (not verified). We give a short self-contained derivation in log-polar coordinates, in which the Hessian is diagonal in each Fourier mode. We restate the threshold as $\cosh^2 r_* = q(N)$ with q rational, and we check the thresholds against an independent 60-digit linear-stability computation for $N = 8, \dots, 15$. We also record an elementary remark: $\xi_*(N)$ is an algebraic unit exactly for $N \in \{8, 9, 11, 15, 23\}$. We also flag an attribution problem in the source corpus: the secondary literature identifies Kimura's 1999 Hamiltonian with the *tanh* (image) model, not the sinh model.

1 Model

Poincaré disk $\mathbb{D} = \{|z| < 1\}$ with metric $4|dz|^2/(1-|z|^2)^2$ (curvature -1) and area (symplectic) form $\omega = \sum_k \lambda(z_k)\, dx_k \wedge dy_k$, $\lambda(z) = 4/(1-|z|^2)^2$. For N unit vortices:

$$H = -\sum_{j<k} \ln|z_j - z_k| + \frac{N-1}{2}\sum_k \ln(1-|z_k|^2) = -\sum_{j<k} \ln\sinh\frac{d_{jk}}{2} + \text{const}, \tag{1}$$

using $\sinh(d/2) = |z-w|/\sqrt{(1-|z|^2)(1-|w|^2)}$. Rotations about 0 are generated by $J = \sum_k (1+|z_k|^2)/(1-|z_k|^2)$ $(= \sum_k \cosh r_k)$. Hamilton's equations are $\dot{x}_k = \lambda^{-1}\partial_{y_k} H$ and $\dot{y}_k = -\lambda^{-1}\partial_{x_k} H$.

Which model this is. Up to the curvature normalisation, (1) is the $\lambda < 0$ member of the family built on the Green's function $\log\big(|z-w|^2/(\sigma_z\sigma_w)\big)$, $\sigma = 1+\lambda|z|^2$, in Montaldi–Tokieda [1, (1.12)]. The source corpus called (1) "Kimura's model". However, Montaldi–Nava-Gaxiola [2, eq. (12)] write Kimura's Hamiltonian as $-\frac{1}{4\pi}\sum \Gamma_i\Gamma_j \ln\frac{\langle X_i,X_j\rangle+1}{\langle X_i,X_j\rangle-1}$ with $\langle X, Y\rangle = -\cosh d$, i.e. $-\frac{1}{4\pi}\sum \ln\tanh^2(d/2)$. Montaldi–Tokieda [1, §4.2] likewise say that the image Green's function $\log|z-w|^2/|1+\lambda z\bar{w}|^2$ "reduces to the Hamiltonian adopted by Kimura". We could not obtain Kimura [3] itself. We therefore call (1) the *sinh model* and make no claim about Kimura's paper.

2 Derivation of the mode decomposition (C1)

Write $z_k = \sqrt{\xi}\, e^{s_k + i(2\pi k/N + \theta_k)}$, so the ring is $s = \theta = 0$. In these coordinates $\omega = \sum_k \kappa(s_k)\, ds_k \wedge d\theta_k$ with $\kappa = 4\xi e^{2s}/(1-\xi e^{2s})^2$, and $\kappa_0 := \kappa(0) = 4\xi/(1-\xi)^2$ at the ring.

Pair term. With $\phi = 2\pi(j-k)/N$, $\Delta s = s_j - s_k$, $\Delta\theta = \theta_j - \theta_k$,

$$\ln|z_j - z_k|^2 = s_j + s_k + \ln\xi + \ln\bigl(2\cosh\Delta s - 2\cos(\phi + \Delta\theta)\bigr).$$

Expanding to second order (sympy, `verify_core.py`) gives no first-order Δs term, no $\Delta s\,\Delta\theta$ cross term, and quadratic part $(\Delta s^2 - \Delta\theta^2)/D_\phi$ with $D_\phi = 2 - 2\cos\phi$. Hence the quadratic part of $H_{\text{pair}} = -\frac{1}{2}\sum_{j<k}\ln|z_j - z_k|^2$ is $-\frac{1}{2}Q(s) + \frac{1}{2}Q(\theta)$, where $Q(x) = \sum_{j<k}(x_j - x_k)^2/D_{jk}$.

Lemma 1. *For $0 \le m \le N$:* $\displaystyle\sum_{l=1}^{N-1} \frac{1-\cos(2\pi ml/N)}{1-\cos(2\pi l/N)} = m(N-m).$

Proof. Let $w = e^{2\pi i l/N}$. For $l \neq 0$, $\frac{1-\cos(2\pi ml/N)}{1-\cos(2\pi l/N)} = \frac{|1-w^m|^2}{|1-w|^2} = \bigl|\sum_{a=0}^{m-1} w^a\bigr|^2 = \sum_{a,b=0}^{m-1} w^{a-b}$. Summing over $l = 0, \dots, N-1$ gives $N \cdot \#\{(a,b) : a \equiv b \bmod N\} = Nm$, because $|a-b| < m \le N$. Subtracting the $l = 0$ term m^2 leaves $m(N-m)$. □

(Equivalent to the identity (2.4) quoted in [1]. Checked numerically to 50 digits for $N \le 80$.) On the Fourier mode $x_k = e^{2\pi imk/N}$, the lemma gives $Q(x) = \frac{m(N-m)}{2}\sum_k |x_k|^2$. So the Hessian of H_{pair} is diagonal on mode m, with entry $-m(N-m)/2$ in s and $+m(N-m)/2$ in θ.

One-body terms. $W(s) = \frac{N-1}{2}\ln(1-\xi e^{2s})$ and $J_1(s) = (1+\xi e^{2s})/(1-\xi e^{2s})$ do not depend on θ. The linear part $-\frac{N-1}{2}\sum s_k$ of H_{pair} together with $W'(0)$ and $-\Omega J_1'(0)$ gives the equilibrium condition, hence

$$\Omega = -\frac{(N-1)(1+\xi)(1-\xi)}{8\xi}, \qquad W''(0) - \Omega J_1''(0) = -\frac{2(N-1)\xi}{(1-\xi)^2} + \frac{(N-1)(1+\xi)^2}{(1-\xi)^2} = \frac{(N-1)(1+\xi^2)}{(1-\xi)^2} =: C_1(\xi).$$

Proposition 2 (C1; known [1])**.** *In the coordinates (s, θ), the Hessian of $F = H - \Omega J$ at the ring is block-diagonal over Fourier modes $m = 0, \dots, N-1$. On mode m it is* $\operatorname{diag}(a_m, b_m)$ *with*

$$a_m = C_1(\xi) - \tfrac{m(N-m)}{2}, \qquad b_m = \tfrac{m(N-m)}{2}.$$

The linearised flow on mode m is $\dot s = (b_m/\kappa_0)\theta$, $\dot\theta = -(a_m/\kappa_0)s$, with eigenvalues $\sigma^2 = -a_m b_m/\kappa_0^2$.

Because $b_m > 0$ for $1 \le m \le N-1$, and $m = 0$ is the rotation direction ($b_0 = 0$), the ring is spectrally stable iff $a_m \ge 0$ for all m. It is linearly unstable (a real pair $\pm\sigma$) iff some $a_m < 0$. Since $\max_m m(N-m)/2 = \lfloor N/2\rfloor\lceil N/2\rceil/2 =: T$, the criterion is $C_1(\xi) \ge T$. When $C_1(\xi) > T$, the Hessian of F is positive definite transverse to the rotation orbit. Montaldi–Tokieda, Theorem 2 (stated for exactly this model), then gives Lyapunov stability modulo rotations. We did not re-prove that energy–momentum step.

3 Marginal stability, the $\cosh^2$ form (C2)

With $\xi = \tanh^2(r/2)$ (r the hyperbolic ring radius), $\cosh r = (1+\xi)/(1-\xi)$ and $(1+\xi^2)/(1-\xi)^2 = (1+\cosh^2 r)/2$. So $C_1 = T$ is equivalent to

$$\cosh^2 r_* = \frac{2T}{N-1} - 1 = q(N), \qquad q(N) = \begin{cases} \frac{N-3}{4} & N \text{ odd}, \\ \frac{(N-2)^2}{4(N-1)} & N \text{ even}, \end{cases} \qquad \xi_* = \frac{\sqrt{q}-1}{\sqrt{q}+1}.$$

$q(N) \le 1$ for $N \le 7$, so every ring is stable ($N = 7$: $q = 1$, $r_* = 0$, the planar Thomson–Havelock degeneracy). For $N \ge 8$ we have $q > 1$, and the ring is stable iff $r > r_*(N)$. This is Montaldi–Tokieda's (2.11) rewritten. Their table of bifurcation values $\lambda r_0^2 = -0.0627, -0.101, -0.143, -0.172, -0.202, -0.225$ ($n = 8, \dots, 13$) is $-\xi_*(N)$, and their example $n = 15$ gives $\xi_* = 2 - \sqrt{3}$.

N	$q(N)$	ξ_*	$\sqrt{\xi_*}$ (Euclid.)	r_* (hyperbolic)
8	9/7	$8 - 3\sqrt{7} = 0.0627460668062\dots$	0.250492	0.511875
9	3/2	$5 - 2\sqrt{6} = 0.1010205144336\dots$	0.317837	0.658479
10	16/9	1/7	0.377964	0.795365
11	2	$3 - 2\sqrt{2} = 0.1715728752538\dots$	0.414214	0.881374
12	25/11	$0.2024108640318\dots$	0.449901	0.969152
13	5/2	$0.2251482265544\dots$	0.474498	1.031719
14	36/13	$0.2492775954100\dots$	0.499277	1.096685
15	3	$2 - \sqrt{3} = 0.2679491924311\dots$	0.517638	1.146216

Numerical check against the dynamics (NUMERICAL). `work/lin_stab.py` codes the Cartesian vector field of (1) directly, without the Fourier reduction. It solves for Ω and forms the full $2N \times 2N$ Jacobian by central differences at 60 digits ($h = 10^{-25}$). (A) For $N = 8, \dots, 15$ and $\xi \in \{0.02,\ 0.9\xi_*,\ 1.1\xi_*,\ 0.5\}$, the spectrum matches $\{0, 0\} \cup \{\pm\sqrt{-a_m b_m}/\kappa_0\}$ to within $2.5 \cdot 10^{-17}$, and Ω matches the formula to 10^{-58}. (B) Bisection on "$\max \mathrm{Re}\,\sigma > 10^{-15}$" locates the threshold within $1.2 \cdot 10^{-19}$ of $\xi_*(N)$ for every $N = 8, \dots, 15$, which is the bisection resolution. Recorded output: `work/lin_stab.out`. The coarser reviewer scans (`checks/phys/scan.py`, sinh branch: first stable $r_E = 0.252, 0.318, 0.378$) agree.

4 An arithmetic remark (C3)

From $\xi_* = (\sqrt{q}-1)/(\sqrt{q}+1)$:

$$\xi_* + \xi_*^{-1} = \frac{2(q+1)}{q-1} = 2 + \frac{4}{q-1} =: c(N), \qquad \frac{4}{q-1} = \begin{cases} \dfrac{16}{N-7} & N \text{ odd}, \\ \dfrac{16(N-1)}{N^2-8N+8} & N \text{ even}. \end{cases}$$

Proposition 3. *For $N \ge 8$ the following are equivalent: (i) $\xi_*(N)$ is an algebraic integer; (ii) $\xi_*(N)$ is an algebraic unit; (iii) $c(N) \in \mathbb{Z}$; (iv) $N \in \{8, 9, 11, 15, 23\}$. In those cases $c = 16, 10, 6, 4, 3$.*

Proof. ξ_* is a root of $x^2 - cx + 1$ and $0 < \xi_* < 1$. (iii)⇒(ii): ξ_* is a root of the monic integer polynomial $x^2 - cx + 1$, and $\xi_*(c - \xi_*) = 1$. (ii)⇒(i) is trivial. (i)⇒(iii): if $\xi_* \in \mathbb{Q}$, it would be a rational algebraic integer, hence in $\mathbb{Z}$, which is impossible in $(0, 1)$. Otherwise its minimal polynomial over $\mathbb{Q}$ is $x^2 - cx + 1$, which has integer coefficients when ξ_* is integral, so $c \in \mathbb{Z}$.

(iii)⇔(iv): for odd N, $16/(N-7) \in \mathbb{Z}$ iff $(N-7) \mid 16$, i.e. $N \in \{9, 11, 15, 23\}$ ($N = 8$ is even). For even $N \geq 8$, $N^2 - 8N + 8 > 0$, and $N^2 - 8N + 8 - 16(N-1) = N^2 - 24N + 24 > 0$ for $N \geq 23$, so $0 < 4/(q-1) < 1$ there. Checking the even $N \in [8, 22]$ leaves only $N = 8$, where $4/(q-1) = 14$. □

Thus $\xi_*(8) = 8 - 3\sqrt{7} = (8 + 3\sqrt{7})^{-1}$, a root of $\xi^2 - 16\xi + 1$. The statement "ξ_* is a unit" is false in general: $\xi_*(10) = 1/7$ has norm 1 (since $\xi_* \cdot \xi_*^{-1} = 1$) but is not integral. Literature: we found no statement of Proposition 3. It is an elementary observation about known numbers and carries no dynamical content. The $SL(2, \mathbb{Z})$ remarks of claim C4 are outside this draft.

5 Remarks on other models (context only)

- *tanh (image) model*, pair term $-\ln\tanh(d/2)$, same area form. The reviewer scans (`checks/phys/scan.py`, `checks/phys/h2.py`) give first stable $r_E \approx 0.26$, 0.336, 0.408 for $N = 8, 9, 10$, which differ from the sinh values. Per [2, 1] this is the Hamiltonian attributed to Kimura. Montaldi–Tokieda call its stability analysis "no longer tractable analytically". A closed form for its thresholds is **open** (claim C5; not addressed here).
- *flat hard-wall disk* (Euclidean area form, images plus the Robin self-image term). The reviewer scans (`checks/disk.py`, `checks/phys/bec.py`) find $N = 7, 8$ rings unstable at every radius tested. Identical Green's functions do not give identical dynamics.
- **Retracted context.** The source corpus claimed "hard-wall BEC = Poincaré disk" and attached these thresholds to BEC experiments. That claim is false (README §3 item 1) and nothing here supports it.

6 Formalisation status

`lean/MathPursuits/Item01_H2Thresholds.lean` (Lean 4.34.1 + Mathlib) compiles with no `sorry`. Its axioms are `propext`, `Classical.choice` and `Quot.sound`. It proves: $\xi + \xi^{-1} = 2 + 4/(q-1)$ for $\xi = (\sqrt{q}-1)/(\sqrt{q}+1)$, $q > 1$; the odd and even reductions of $4/(q-1)$; $n/d \in \mathbb{Z} \iff d \mid n$; $(N-7) \mid 16 \iff N \in \{9, 11, 15, 23\}$ (odd $N > 7$); $(N^2 - 8N + 8) \mid 16(N-1) \iff N = 8$ (even $N \geq 8$); $c(N) \in \mathbb{Z} \iff N \in \{8, 9, 11, 15, 23\}$ for all $N > 7$, with the values $16, 10, 6, 4, 3$; integrality and unit property of ξ when $c \in \mathbb{Z}$; $\xi_*(8) = 8 - 3\sqrt{7}$ and $\xi_*(8)^2 - 16\xi_*(8) + 1 = 0$; $\xi_*(10) = 1/7$ and $1/7$ not integral over $\mathbb{Z}$. Not formalised: the dynamics (Proposition 2), Lemma 1, and the converse (i)⇒(iii) for irrational ξ_*.

7 Status

C1, C2: **KNOWN** (Montaldi–Tokieda 2013), re-derived and checked (symbolic and 60-digit numerics). C3: elementary, proved, and formalised for its arithmetic part. Novelty POSSIBLY-NEW but immaterial. Not yet reviewed by a second reader. Open literature gaps: Boatto 2008 (text unreadable), Hwang–Kim 2009/2013 and Kimura 1999 (paywalled).

References

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[4] S. Boatto, Curvature perturbations and stability of a ring of vortices, DCDS-B 10 (2008) 349–375 (not read in full).

[5] S. Hwang, S.-C. Kim, Point vortices on hyperbolic sphere, J. Geom. Phys. 59 (2009) 475–488; Relative equilibria of point vortices on the hyperbolic sphere, J. Math. Phys. 54 (2013) 063101 (not obtained).